



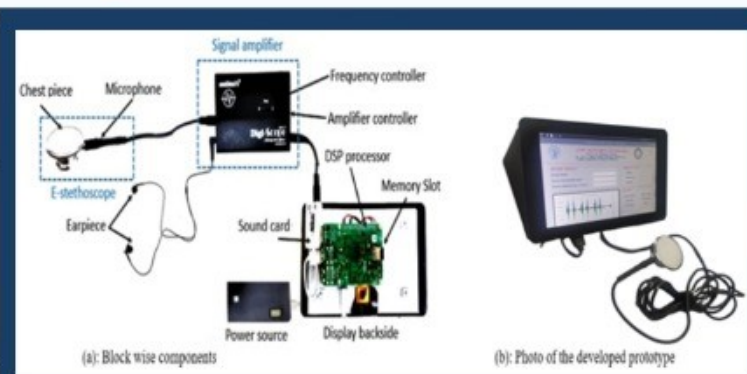
CENTRE FOR ADVANCED STUDIES

LUCKNOW, UTTAR PRADESH, INDIA

INSPIRING STUDENT INNOVATION

“Intelligent Stethoscope” : Artificial Intelligence based Device for Early Medical Diagnosis of Heart, Lung and Prenatal Health

A team of Centre for Advanced Studies developed an Artificial Intelligence based automatic intelligent stethoscope to diagnose cardiac, pulmonary and prenatal disorders to minimize human intervention. The proposed system involves a convolutional neural network (1D-CNN) model to diagnose the cardiac disorders from the PCG signal and lungs disorder from Pulmonary signals because of its high accuracy and robustness. The proposed model is ported into small computing devices to make a stand-alone diagnostic tool that may be of help in remote primary health care centers. The developed prototype contains a stethoscope, sound card, display screen, DSP processor, powerbank. The proposed method is non-invasive, efficient, robust, and has low time complexity making it suitable for real-time applications.



THIS INNOVATION OF CENTRE FOR ADVANCED STUDIES CAN MAKE A SIGNIFICANT CONTRIBUTION FOR THE DIAGNOSIS OF CARDIAC, PULMONARY AND PRENATAL DISORDERS WHICH CAN BE USEFUL FOR SCREENING PURPOSE IN PRIMARY HEALTH CARE CENTRES AND REMOTE AREAS.

RESEARCH TEAM



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PATENTS FROM THE PROJECT

- “A System and Method for Machine Learning Based Automatic Detection of Abnormality from Cardiac Sound” Application number: 01911033584 Dated: 21.08.2019, Status: Under Examination.
- “Artificial Intelligence based System and Method for Multi-Class Classification of Heart Diseases”, Application number 201911047767 Dated: 22.11.2019, Status: Under Examination.
- “Automatic Detection of Lungs Disease using Machine Learning from Respiratory Sound” Application number: 202011009043 Dated: 03.03.2020, Status: Under Examination.

RESEARCH PUBLICATION FROM THIS WORK

1. Anjali Yadav, Anushikha Singh, Malay Kishore Dutta & Carlos M. Travieso “Machine Learning Based Classification of Cardiac Diseases from PCG recorded Heart Sounds” Neural Computing and Applications, Springer Nature Publishers, DOI - 10.1007/s00521-019-04547-5, December 2020, SCI indexed Impact Factor - 4.774.2.
2. Neeraj Baghel, Malay Kishore Dutta & Radim Burget, “Automatic Diagnosis of Multiple Cardiac Diseases from PCG Signals using Convolutional Neural Network” Computer Methods and Programs in Biomedicine, Elsevier Publishers, Volume 197, doi.org/10.1016/j.cmpb.2020.105750, Elsevier Publishers, 2020, SCI indexed, Impact Factor - 3.632.